How the specimens were tested

SampleKit's demo: a tensile test.

Three flat specimens, one of aluminium, one of copper and one of steel, were each pulled until they broke. This note says how, so that a sample can link to it: the method is on page 2.

The method

1. Each specimen is a strip cut from a 2 mm sheet, 12.5 mm wide, with a gauge length of 50 mm.

2. Its width and its thickness are each read three times with a micrometer; the area of its cross-section is their means' product.

3. It is pulled at 2 mm/min, at room temperature, along or across the rolling direction, until it breaks. The machine writes the force read at each strain to a file, curve.csv.

4. The stress is the force divided by the area. The Young's modulus is the slope of the curve's straight start; the yield strength, Rp0.2, is read at 0.2 % of plastic strain; the tensile strength, Rm, is the highest stress; the elongation, A, is the strain at break.

The test follows ISO 6892-1, tensile testing of metallic materials at room temperature.